INTRODUCTION TO IMAGE PROCESSING

LECTURE 1
INTRODUCTION
IMAGE BASICS

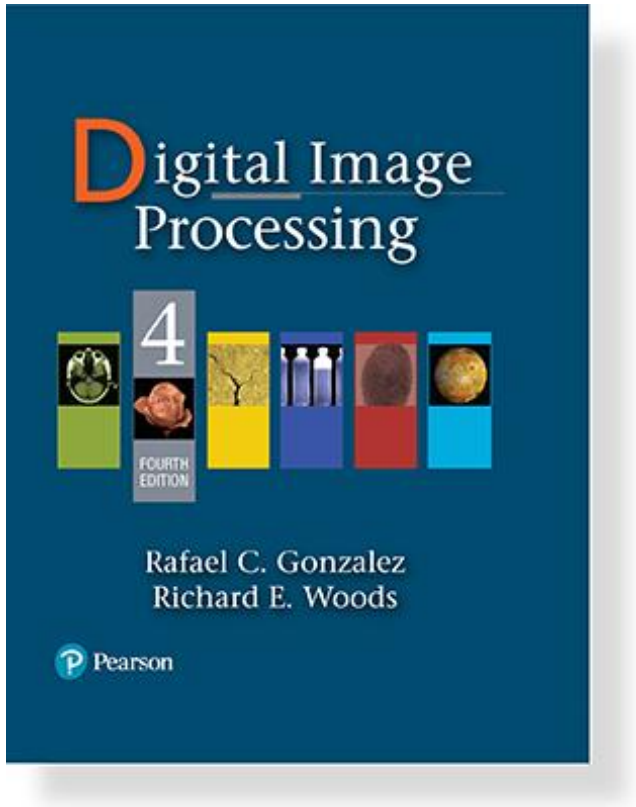
LECTURE RULES

FOLLOW ANNOUNCEMENTS ON CANVAS AT ESUZEM

LECTURE RULES

- **MIDTERM – 40%**
- **HOMEWORKS – 10%**
  (1 to 2 programming homeworks in MATLAB or Python)
- **FINAL – 50%**

RECOMMENDED TEXTBOOK



Digital Image Processing
4th Ed.

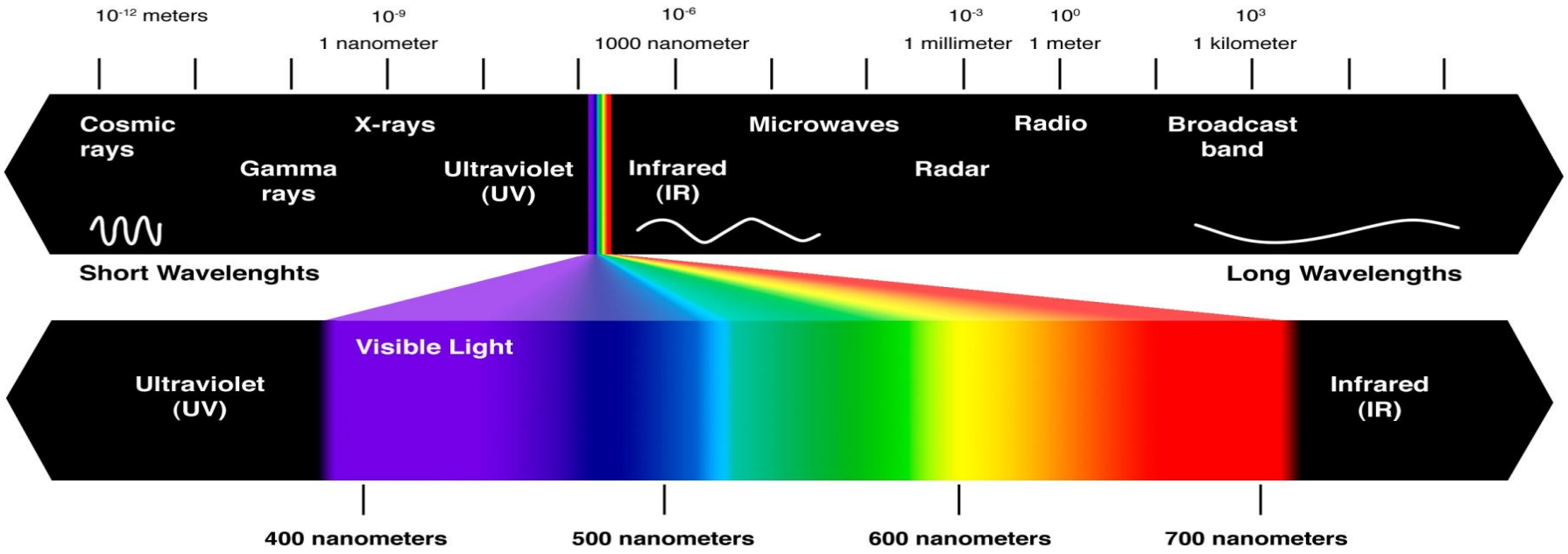
Gonzalez and Woods

© 2018

THE SPECTRUM

Imaging systems cover almost the entire electromagnetic spectrum, ranging from gamma to radio waves.

They can operate on images generated by sources that humans are not used to associate with images.



THE SPECTRUM

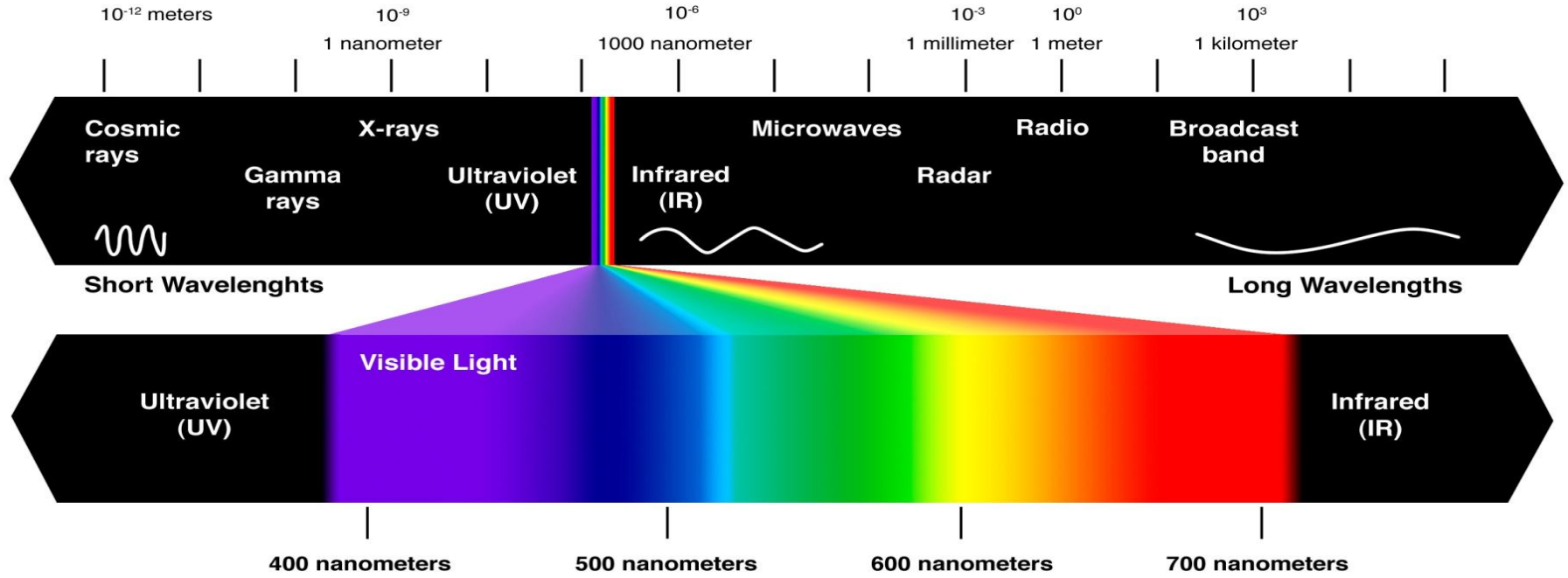
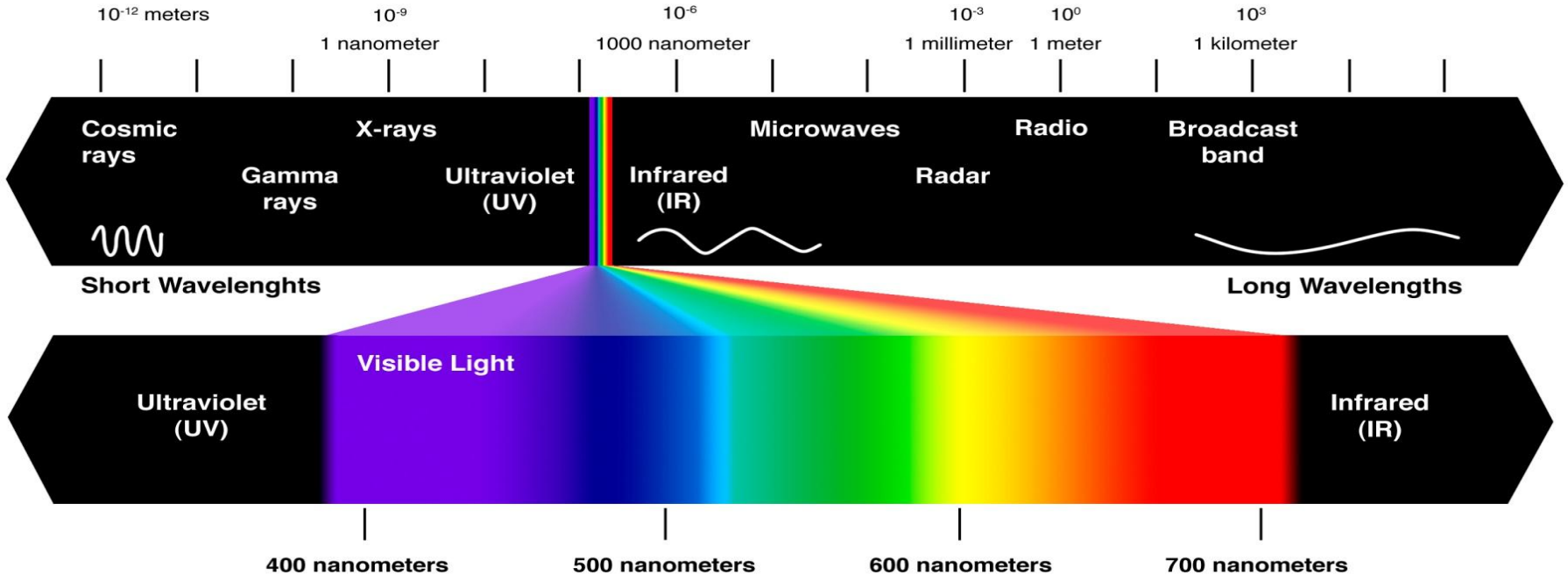


Image analysis is not limited to the visual band of the electromagnetic *spectrum.*

Gamma-ray imaging: nuclear medicine and astronomical observations
X-rays: medical diagnostics, industry, and astronomy, etc.
Ultraviolet: industrial inspection, microscopy, lasers, biological imaging, astronomical observations
Visible and infrared bands: light microscopy, astronomy, remote sensing, industry, entertainment, multimedia, machine vision
Microwave band: radar
Radio band: medicine (such as MRI) and astronomy

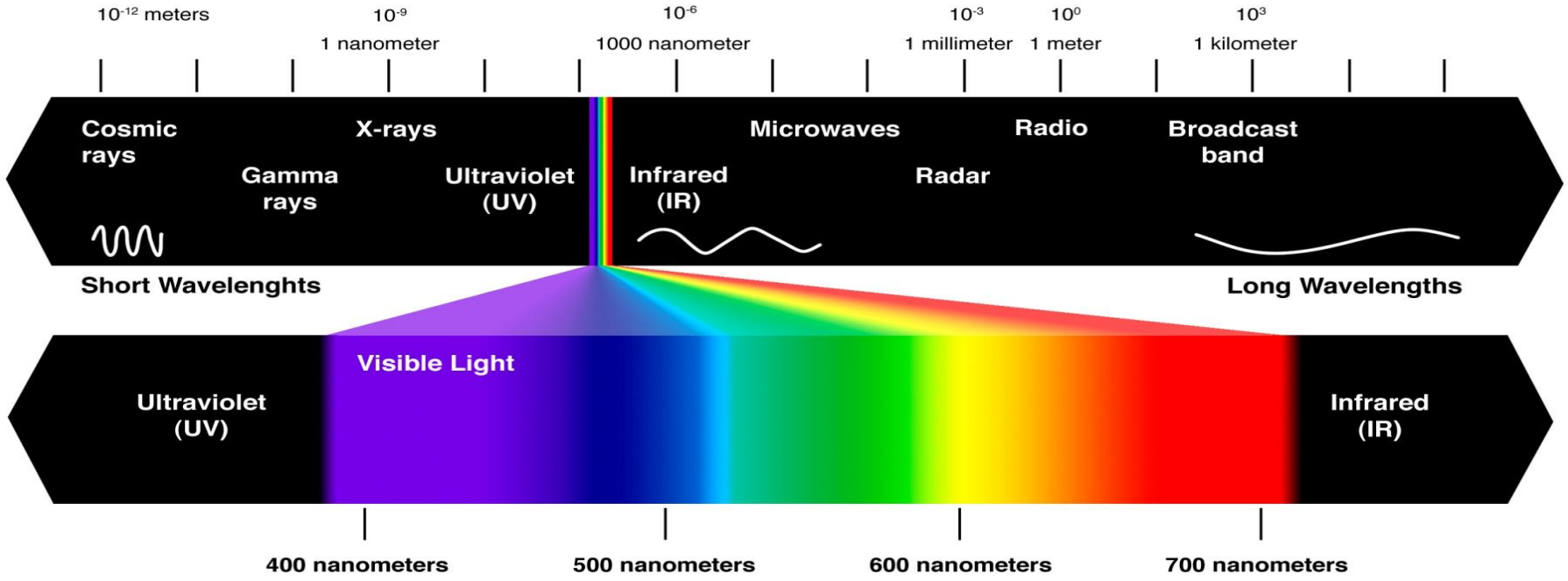
THE SPECTRUM



Different imaging systems measure different physical phenomena, but eventually they can all produce digital images that we can process computationally.

Physical phenomenon → Sensor → Electrical signal → Digital image → Image processing

THE SPECTRUM

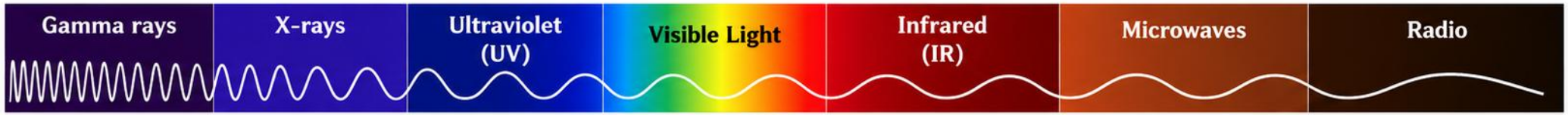


$$c = \lambda f$$

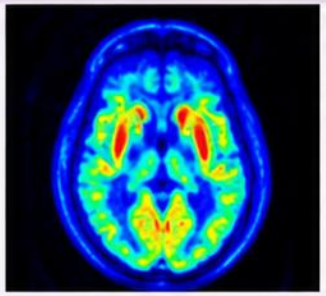
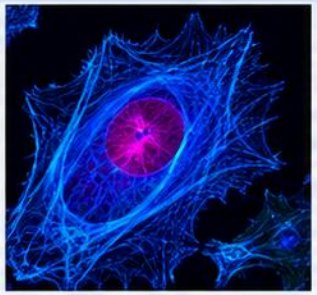
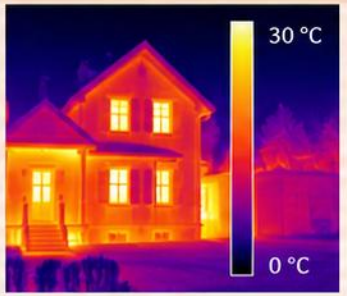
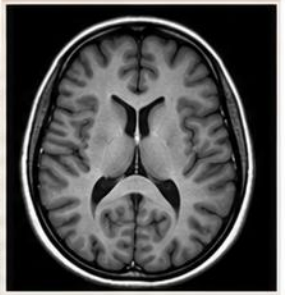
$$E = hf = \frac{hc}{\lambda}$$

$$c = 3 \times 10^8 \text{ m/s}$$

Electromagnetic Spectrum and Example Images



Wavelength (λ)	10^{-12} m	10^{-10} m	10^{-8} m	10^{-6} m	10^{-4} m	10^{-2} m	10^{0} m	10^{2} m
Frequency (f)	10^{20} Hz	10^{18} Hz	10^{16} Hz	10^{14} Hz	10^{12} Hz	10^{10} Hz	10^{8} Hz	10^{6} Hz
Photon energy (E)	10^{6} eV	10^{4} eV	10^{2} eV	1 eV	10^{-2} eV	10^{-4} eV	10^{-6} eV	10^{-8} eV

Gamma-ray imaging	X-ray imaging	Ultraviolet (UV) imaging	Visible Light imaging	Infrared (IR) imaging	Microwave imaging (Radar)	Radio imaging (MRI)
						
PET scan (human brain)	Chest X-ray (medical imaging)	Fluorescence microscopy (cell image)	Color photograph (RGB image)	Thermal image (building)	SAR image (satellite radar)	MRI image (human brain)
Typical applications: Nuclear medicine, astronomy	**Typical applications:** Medical diagnostics, industrial inspection, astronomy	**Typical applications:** Industrial inspection, microscopy, biological imaging, astronomy	**Typical applications:** Photography, machine vision, microscopy, astronomy, multimedia	**Typical applications:** Thermal imaging, remote sensing, industry, surveillance	**Typical applications:** Radar, remote sensing, weather monitoring	**Typical applications:** Medicine (such as MRI) and astronomy

Images







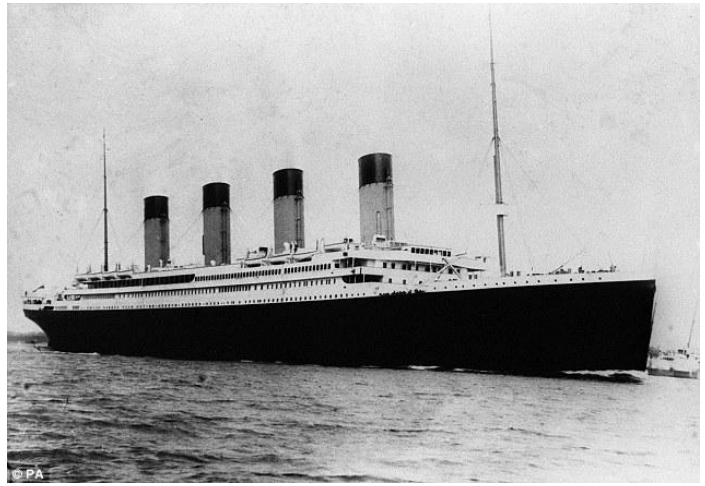
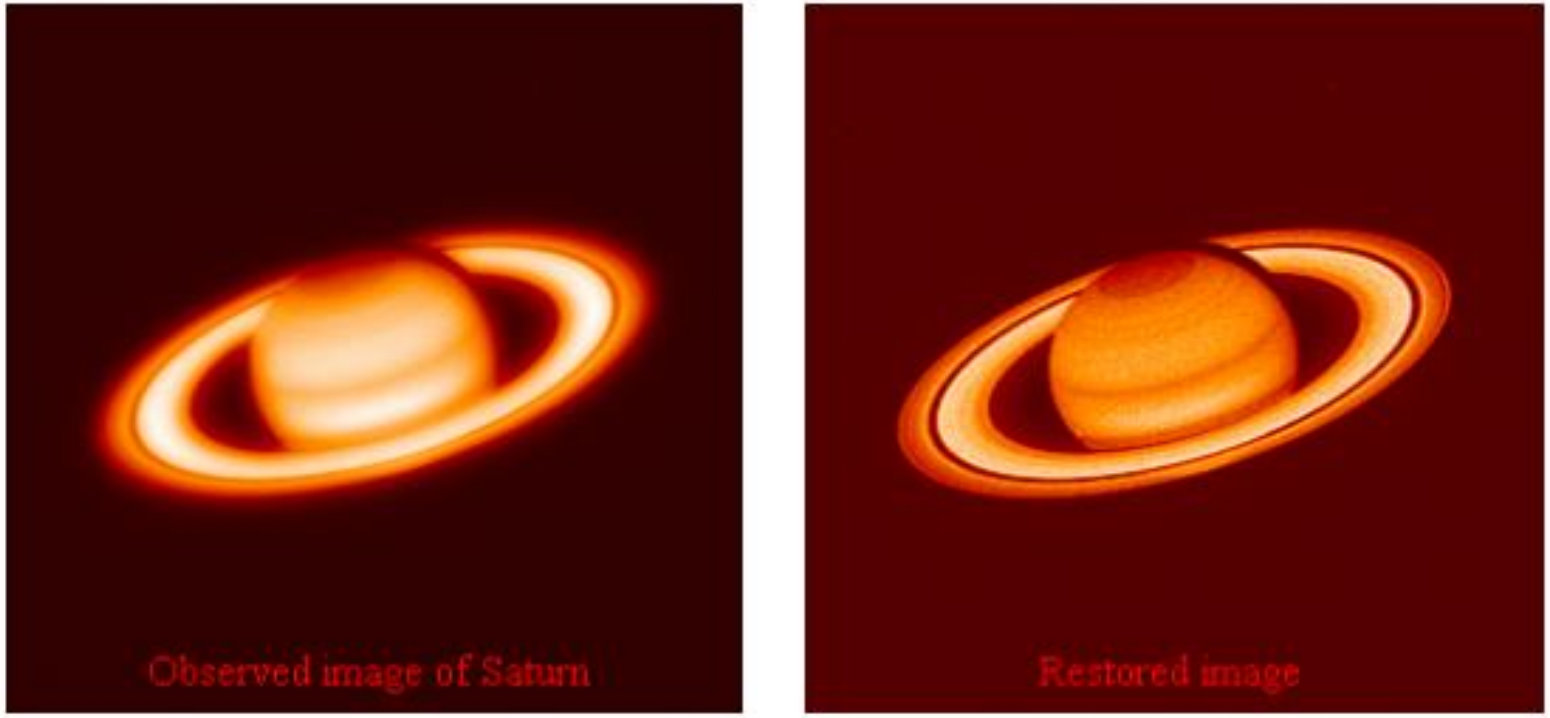






Image Processing Examples

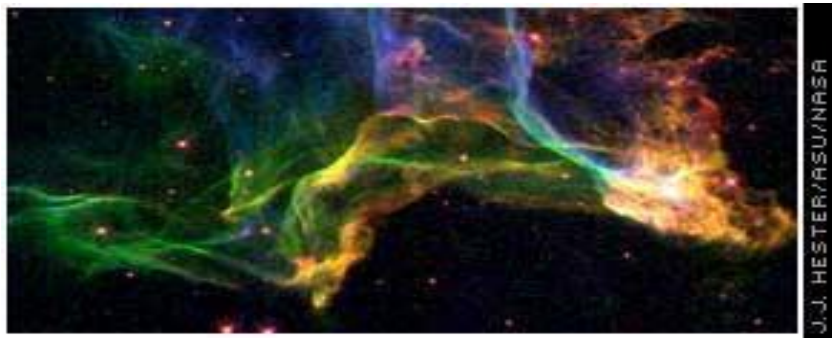
Restoration of image from Hubble Space Telescope

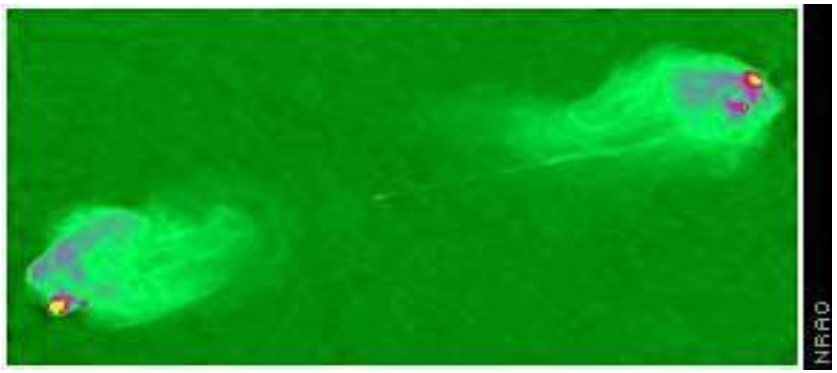


Source: IVPL Northwestern University, Chicago

Image Processing Examples

Gamma-Ray Imaging (freq: > 10^{19} Hz, wavelength: <10^{-12} m)





Cygnus loop, part of a supernova, seen in visible and in an image produced from gamma radiation

Image Processing Examples

Color and intensity enhancement of photographs



Original



Processed

Image Processing Examples

Noise reduction

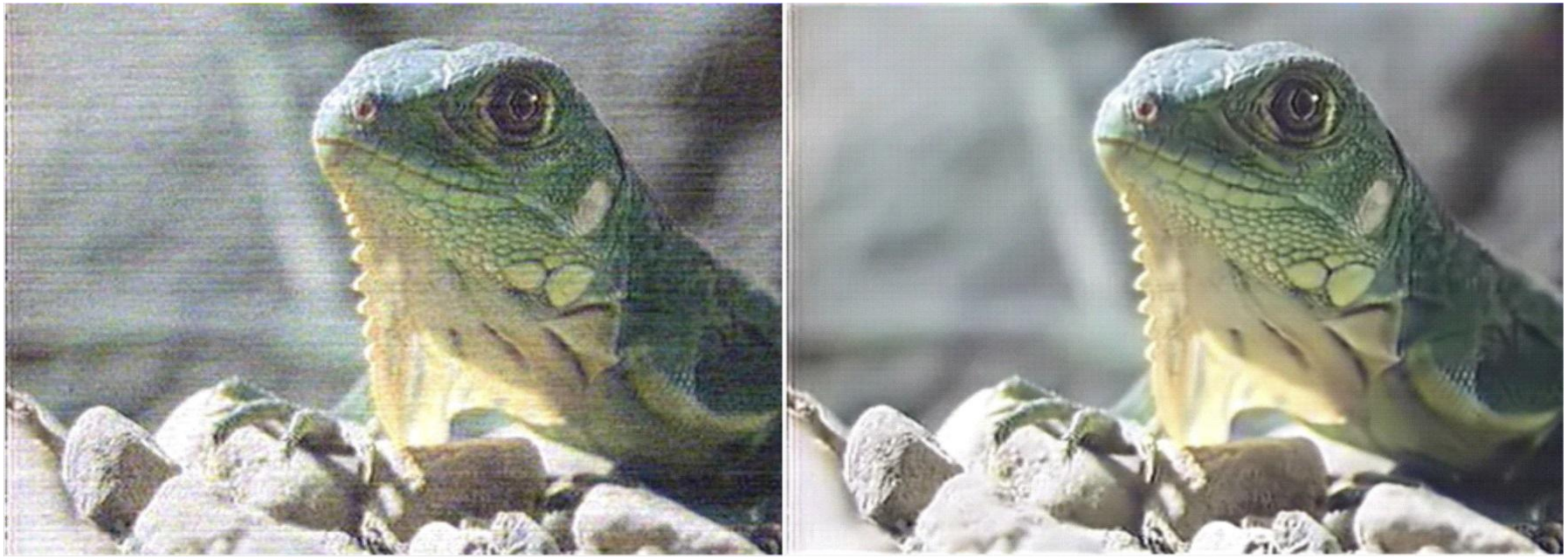


Image Processing Examples

Storing Images

Standard DVD



Blue-ray DVD

Image Processing Examples

Transmitting Images



Video conferencing

Image Processing Examples

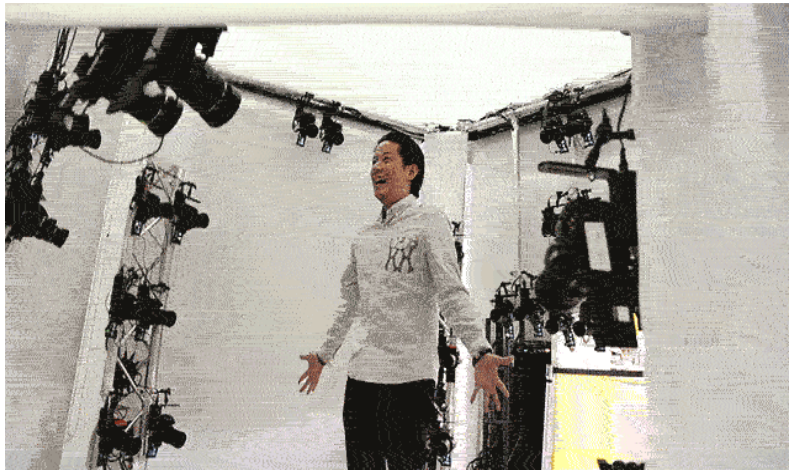
Face morphing



Source: Yi-Wen Liu and Yu-Li Hsueh, EE368 class project, spring 2000.

Image Processing Examples







http://gizmodo.com/nikon-just-put-me-in-a-video-game-and-it-was-totally-in-1497441443

Image Processing Examples

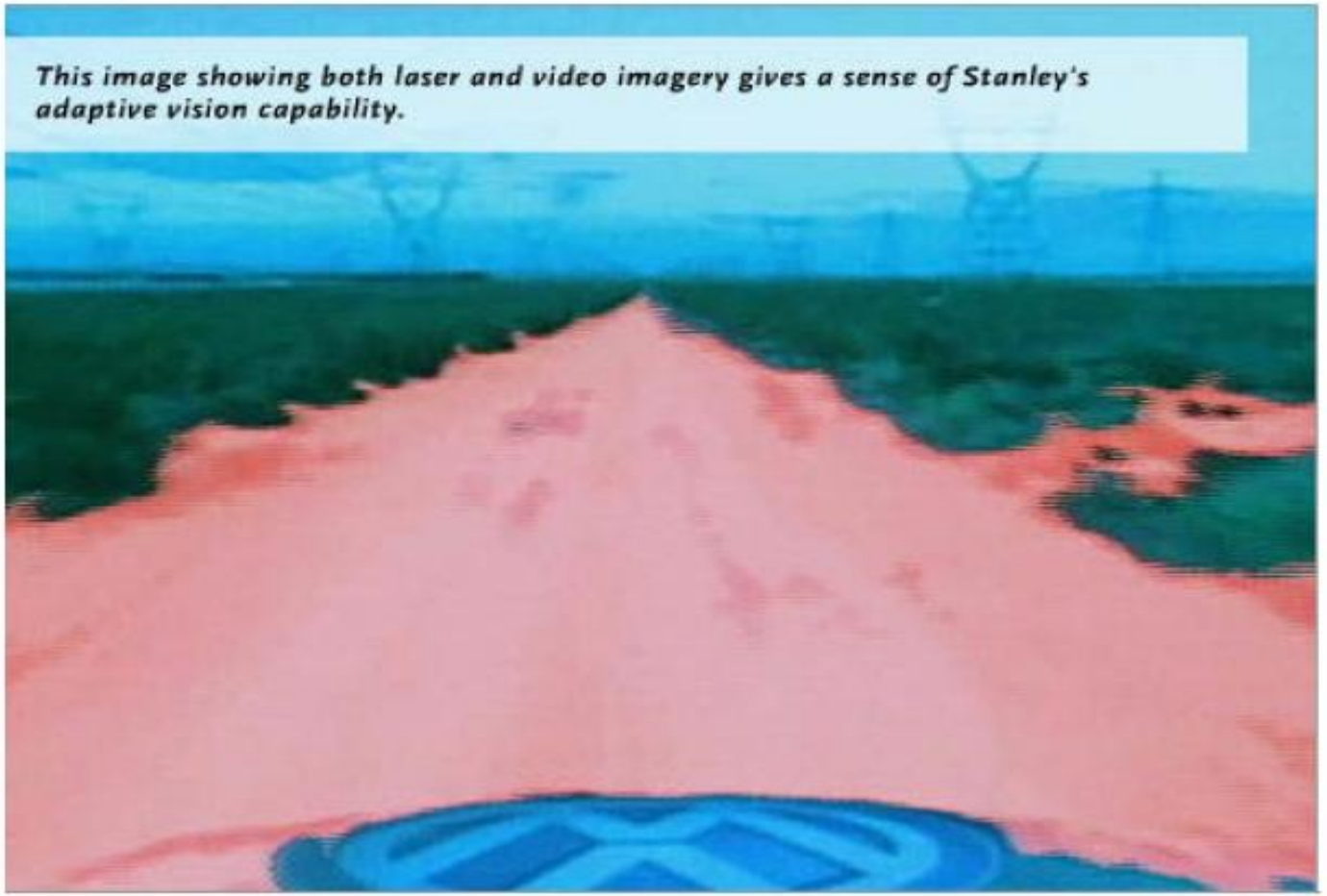
Face Detection



source: Henry Chang, Ulises Robles, EE368 class project, spring 2000.

Example images with more faces
Challenges
False positive
False negative
Compact cameras have face detectors for better focus
Red-eye removal
Human recognition
Human computer interaction

Image Processing Examples





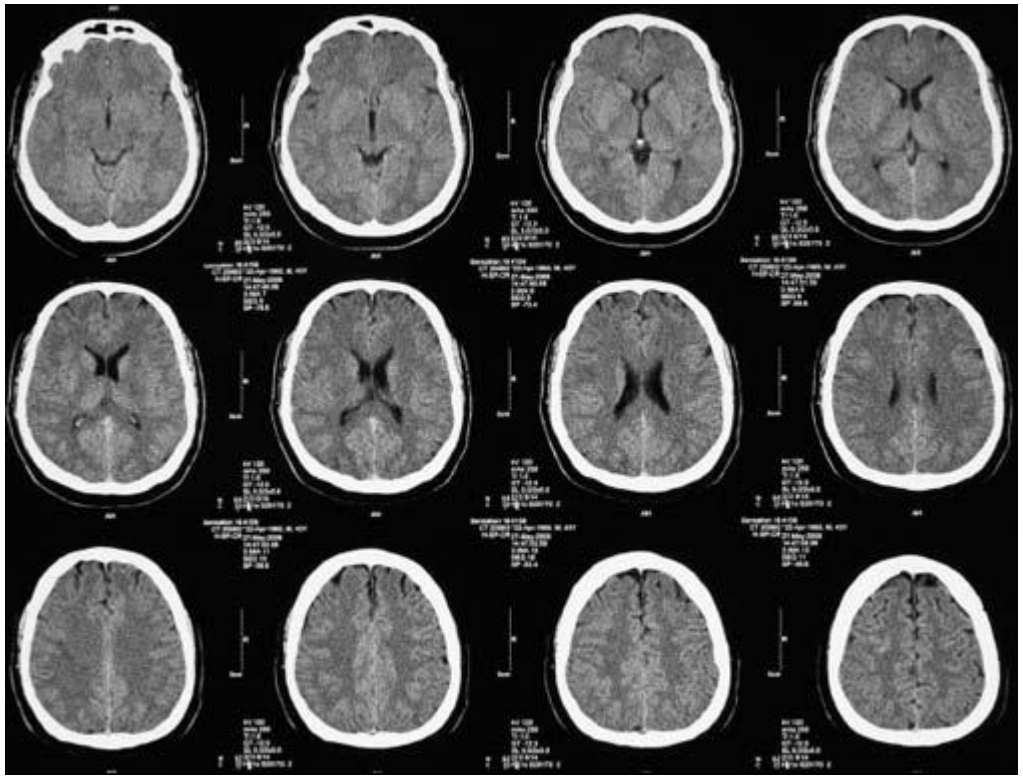
http://cs.stanford.edu/group/roadrunner/stanley.html

Autonomous driving

Image Processing Examples



Image Processing Examples



Computed Tomography
A computed tomography, or CT, scanner uses X-rays to create an image of both bone and brain tissue.

Image Processing Examples

From a 3D MRI scan extract the brain regions and form a 3D model of the brain
Separate from the skull, sinuses, other head regions...

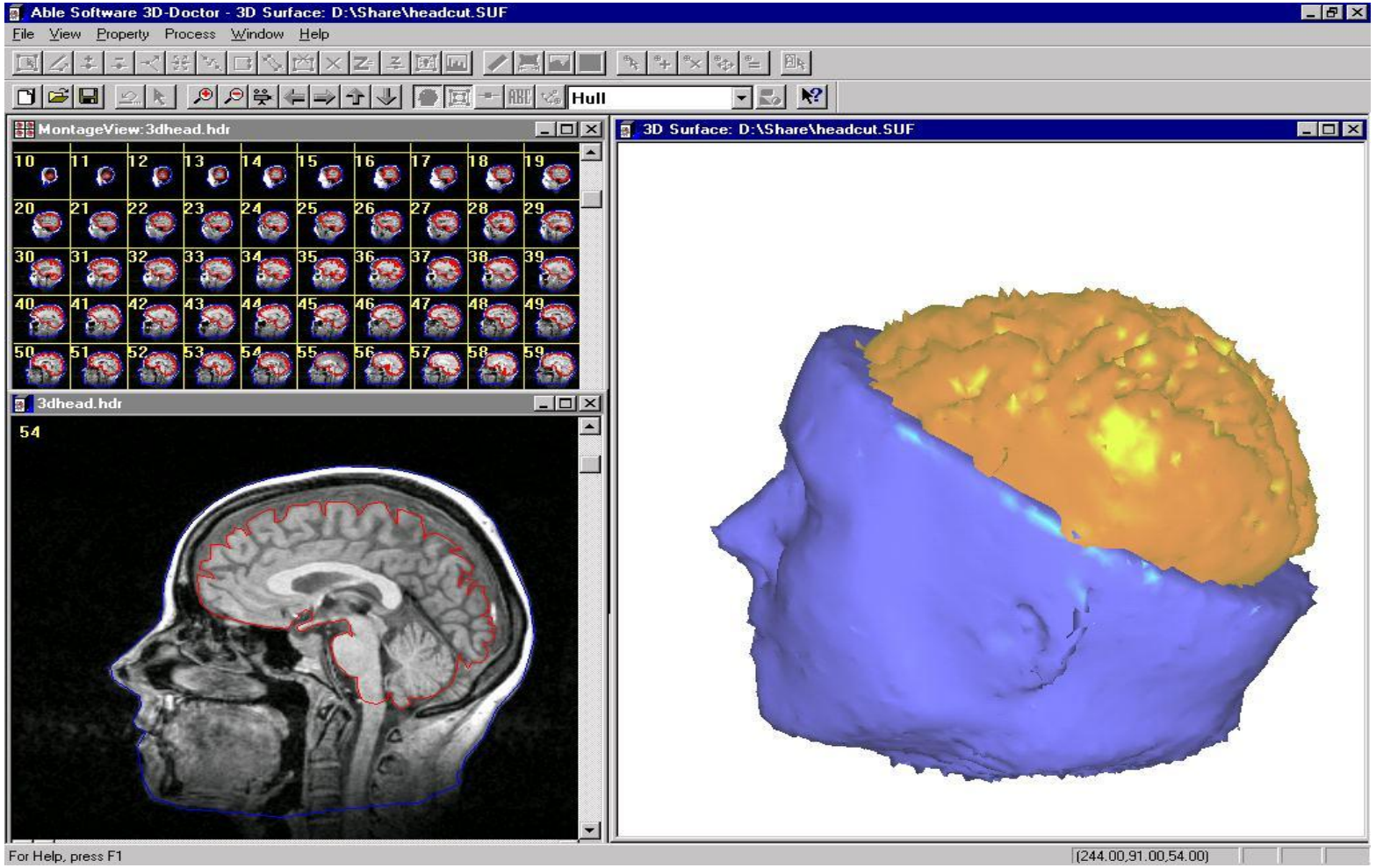


Image Processing Examples

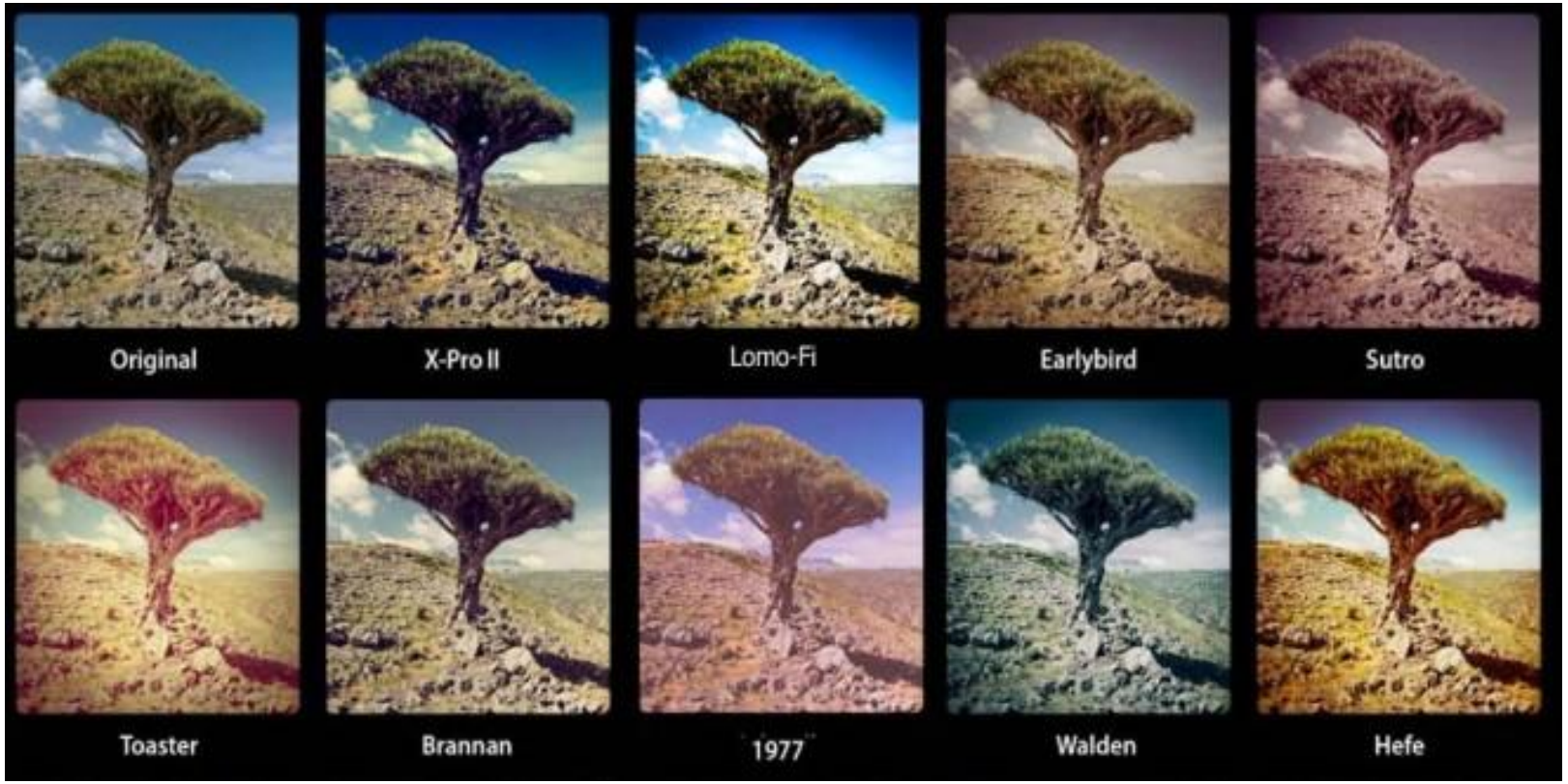
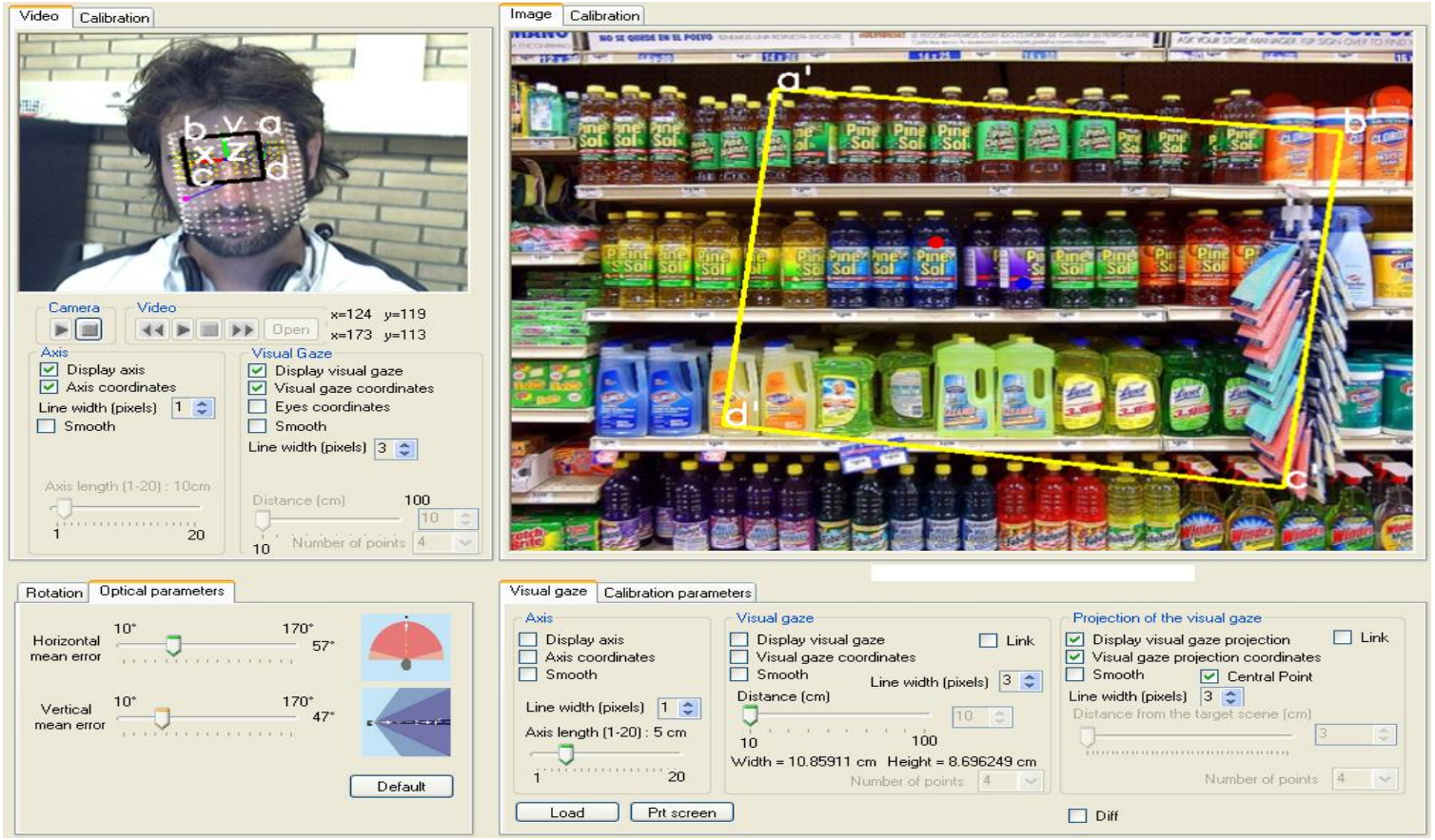


Image Processing Examples



Visual Gaze Estimation by Joint Head and Eye Information

R. Valenti, N. Sebe, T. Gevers

In International Conference on Pattern Recognition 2010.

Image Processing Examples



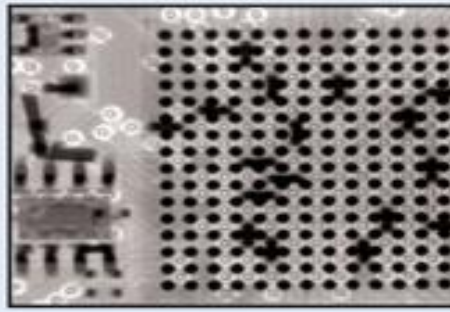
Recognizing facial expressions

Image Processing Examples

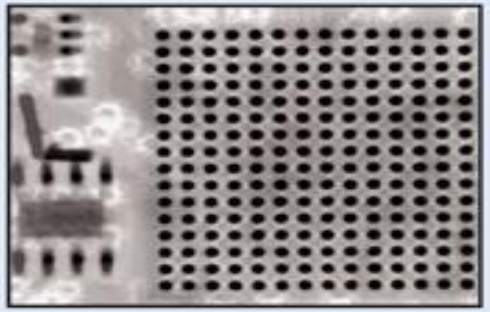


X-Ray Imaging

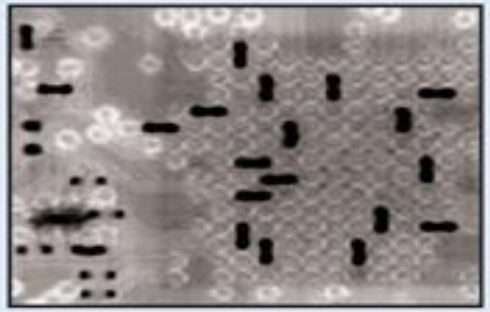
X ray inspection machine for circuit boards



Traditional 2D x-ray image of BGA on a double sided board.

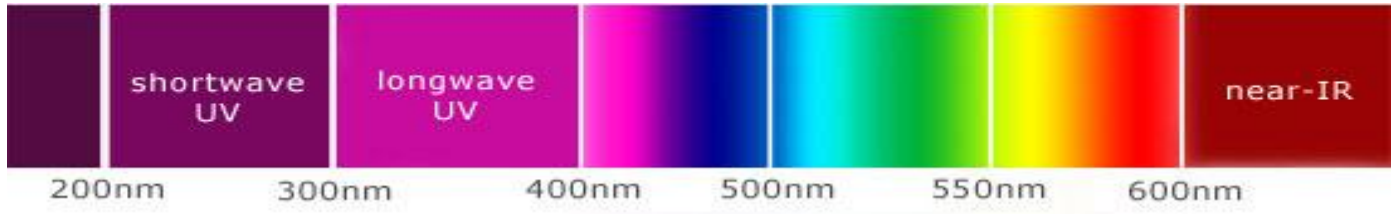


3D x-ray imaging solder joints on top side on a board.



3D x-ray imaging solder joints on bottom side on a board.

Image Processing Examples

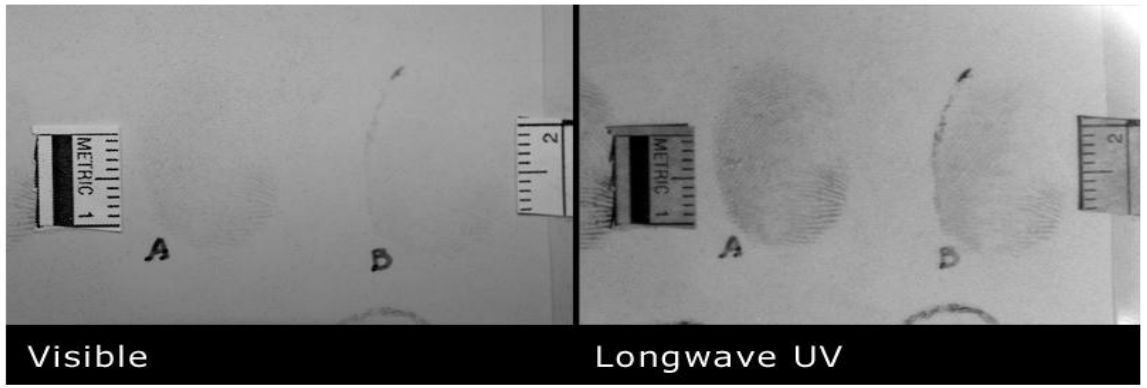


Forensics Imaging Applications

Ultraviolet Imaging



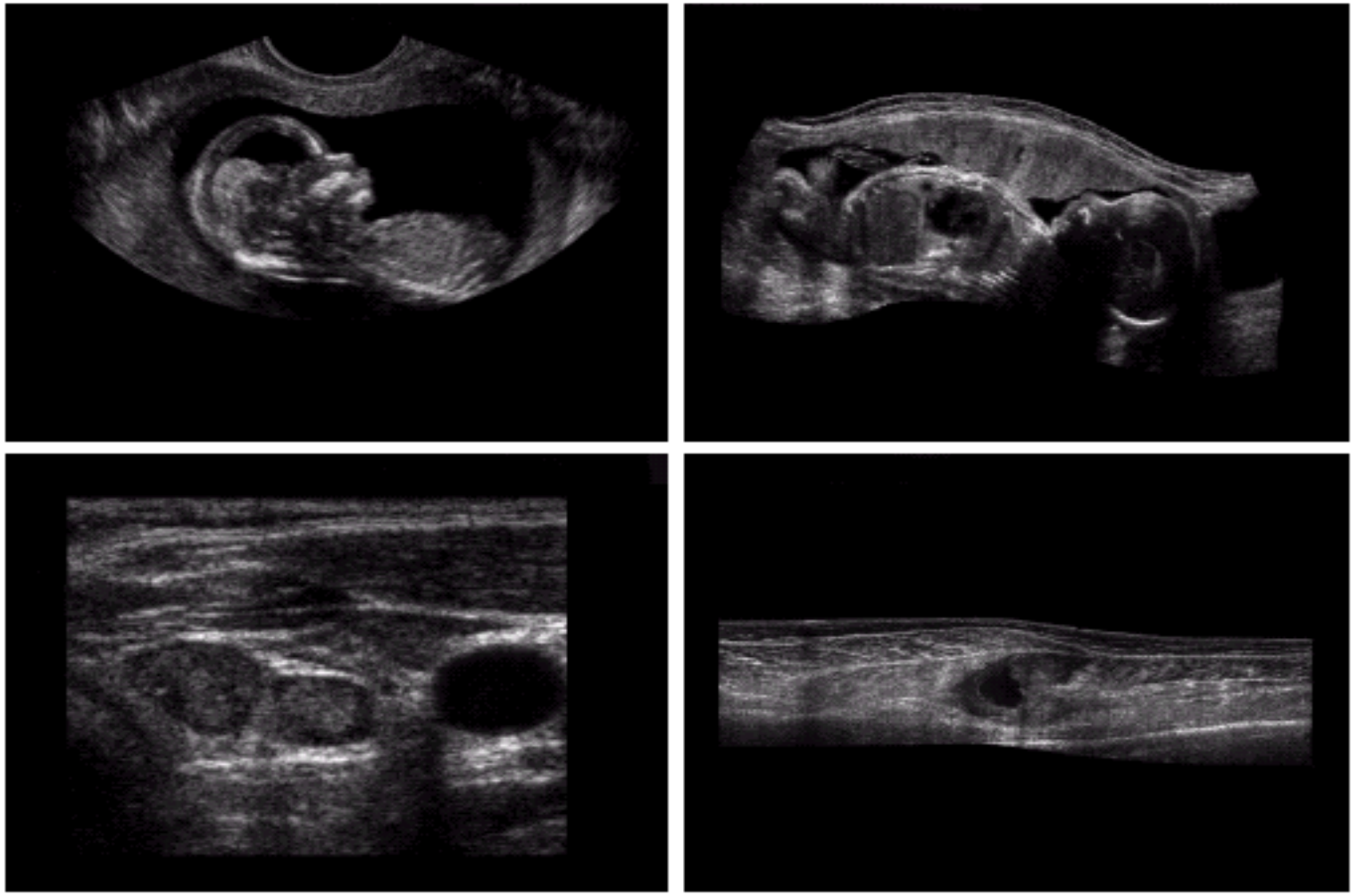
Cleaning marks on ceiling (where blood was cleaned off). The cleaning products leave a slight residue that absorbs UV - © Dr. Austin Richards



Bloody finger prints - Courtesy of Rachel Leintz, Phoenix Police Department, Homicide Division

Image Processing Examples

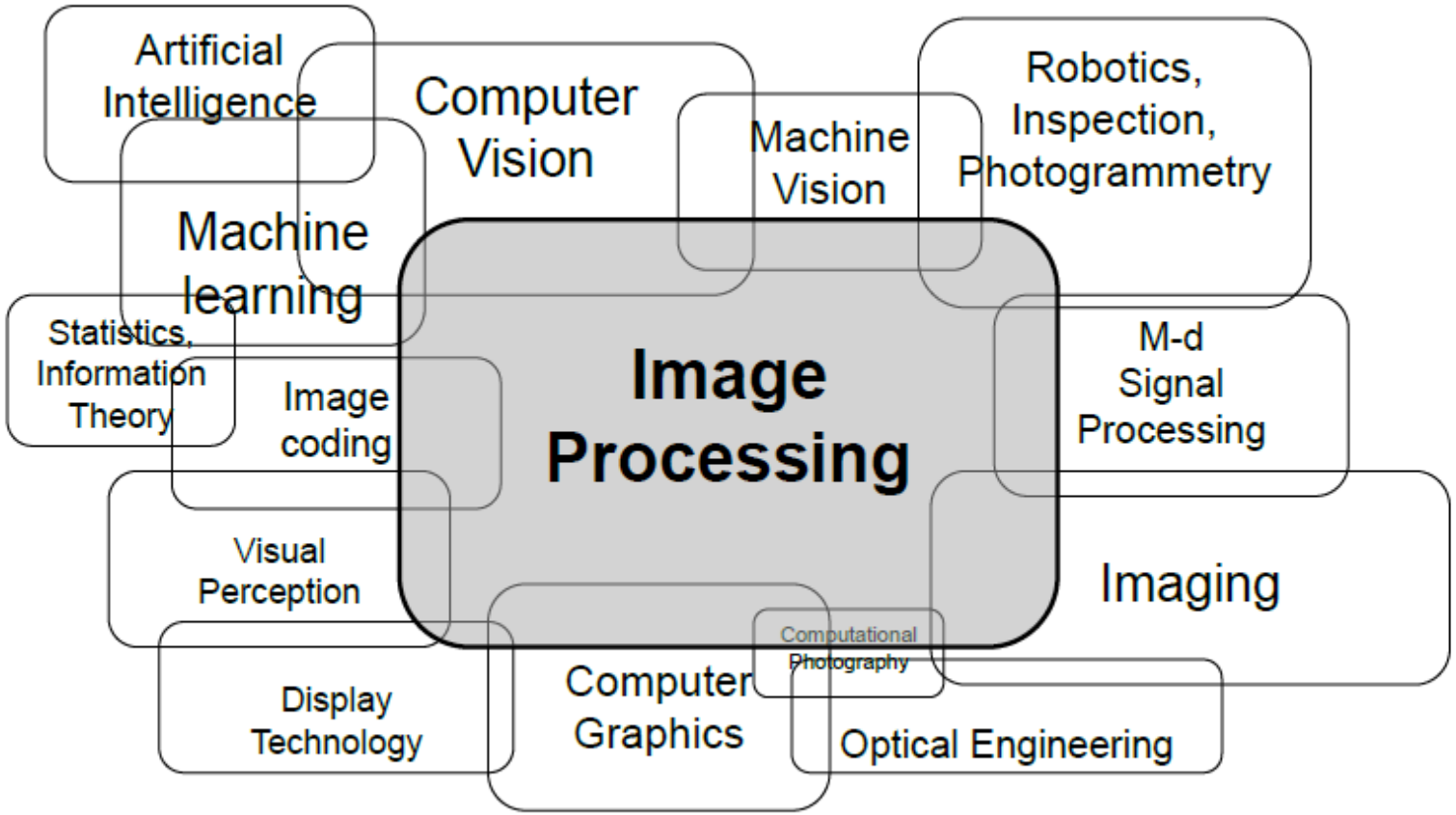
Ultrasound Imaging



a b
c d

FIGURE 1.20 Examples of ultrasound imaging. (a) Baby. (2) Another view of baby. (c) Thyroids. (d) Muscle layers showing lesion. (Courtesy of Siemens Medical Systems, Inc., Ultrasound Group.)

Image Processing and Related Fields



One useful paradigm is to consider three levels of processing:

Low-level processing: Primitive operations such as image preprocessing to reduce noise, contrast enhancement, and image sharpening. Both input and output are images.

Mid-level processing: Processing on images for tasks such as segmentation (partitioning an image into regions or objects), and classification (recognition) of individual objects. The inputs generally are images, the outputs are attributes extracted from those images (e.g., edges, contours, and the identity of individual objects).

High-level processing: “making sense” of an ensemble of recognized objects, as in image analysis, performing the cognitive functions normally associated with vision.

These processes can be performed separately or by a single «end-to-end» method.

Terminology

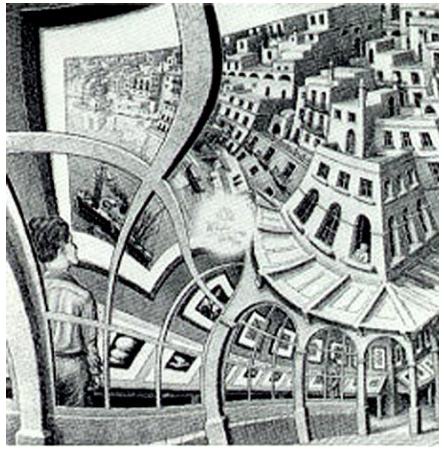
All these fields are very much related:

- **Image processing**
- **Image analysis**
- **Image manipulation**

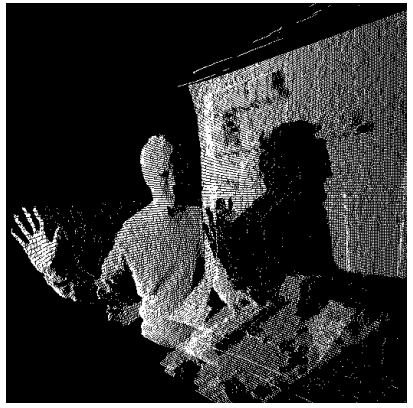
- **Computer Vision**
- **3D Vision**
- **Machine Vision**
- **Robot Vision**
- **Image understanding**

What is an image?

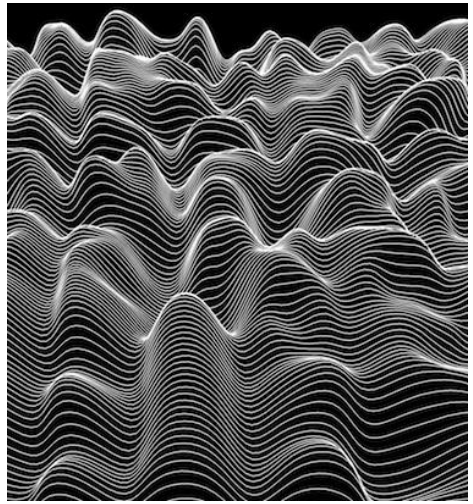


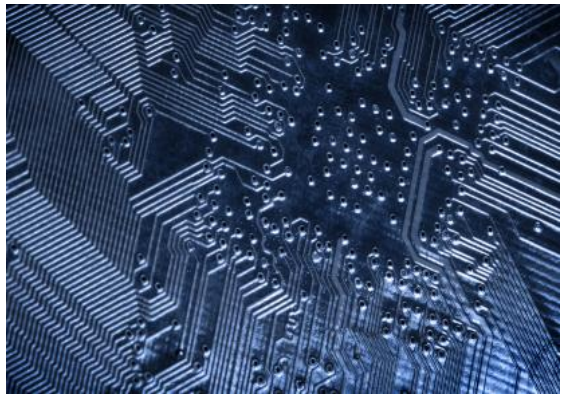














What is an image?

An image may be defined as a two-dimensional function: $f(x, y)$.

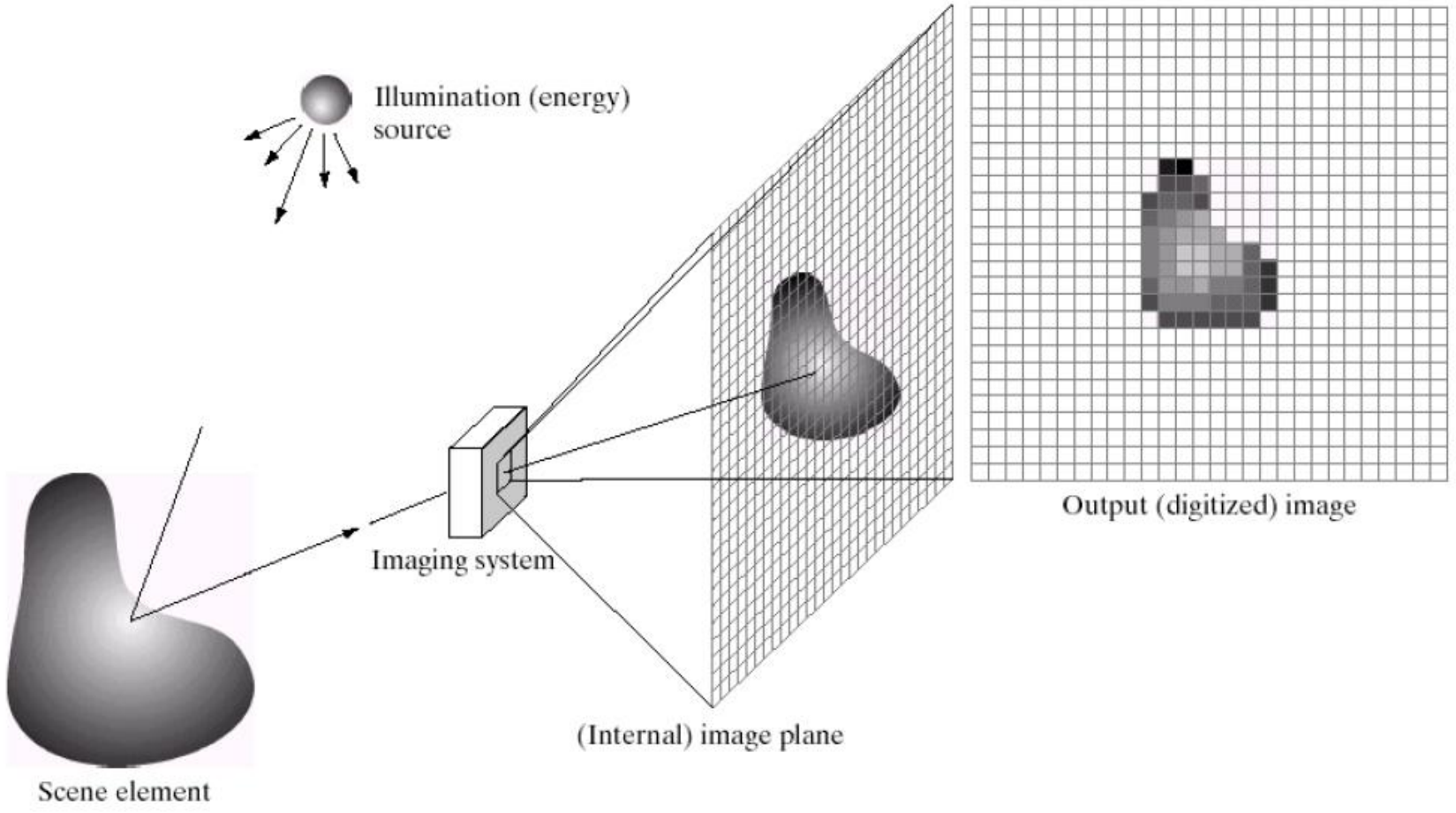
x and y are *spatial* (plane) coordinates.

The amplitude of f at any pair of coordinates (x, y) is called the *intensity* or *gray level* of the image at that point.

When x, y, and the amplitude values of f are all finite, discrete quantities, we call the image a *digital image*.

Digital 2D image

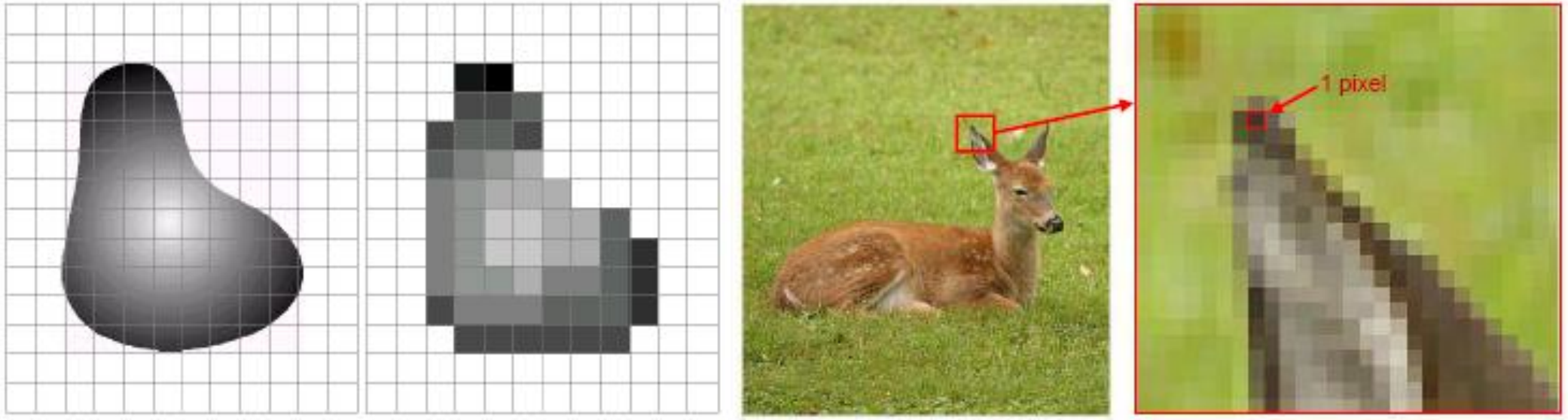
A **digital image** is a representation of a two-dimensional image as a finite set of digital values, called picture elements or **pixels.**



Digital 2D image

A **digital image** is a representation of a two-dimensional image as a finite set of digital values, called picture elements or **pixels.**

Each pixel represents a «location» and a «value».



(a) (b) (c) (d)

(a) Continuos image projected onto a sensor array. (b) Result of image sampling and quantization.

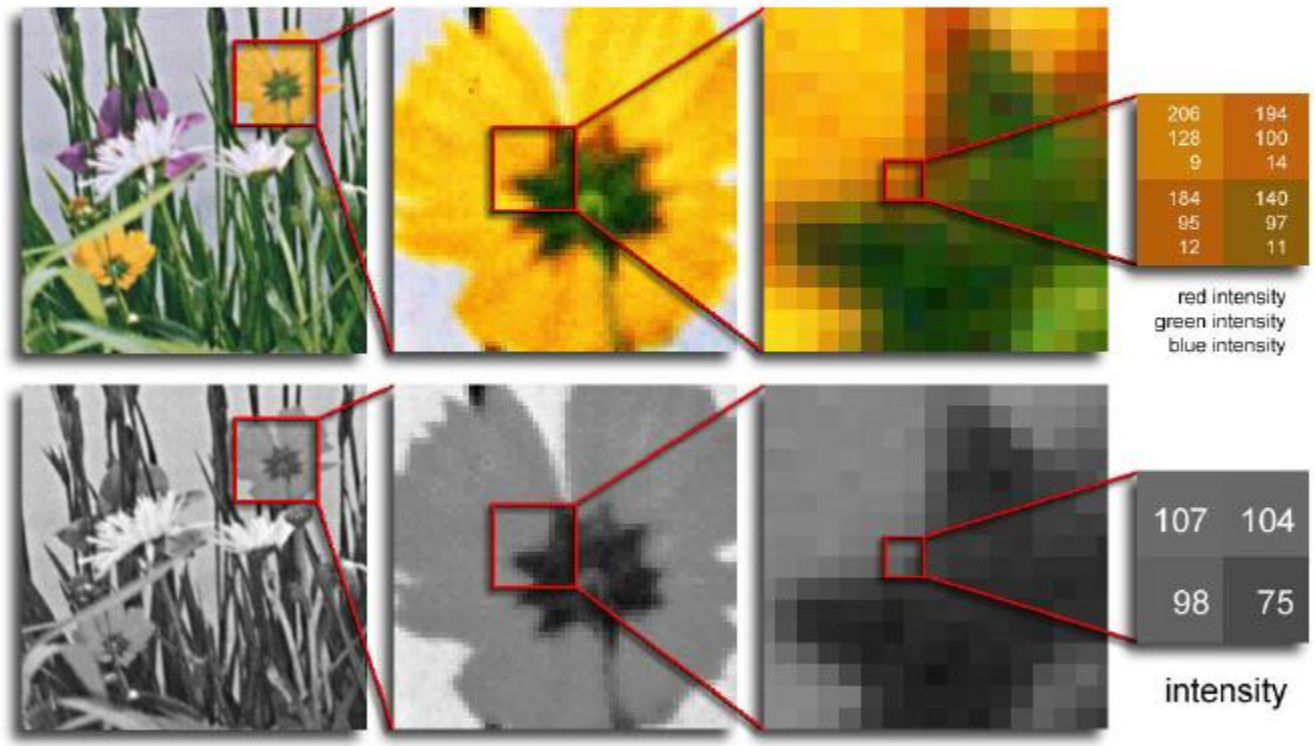
(c) A digital color image (d) Zoomed in

Digital Image

Color images have 3 values per pixel; monochrome images have 1 value per pixel.

a grid of squares, each of which contains a single color

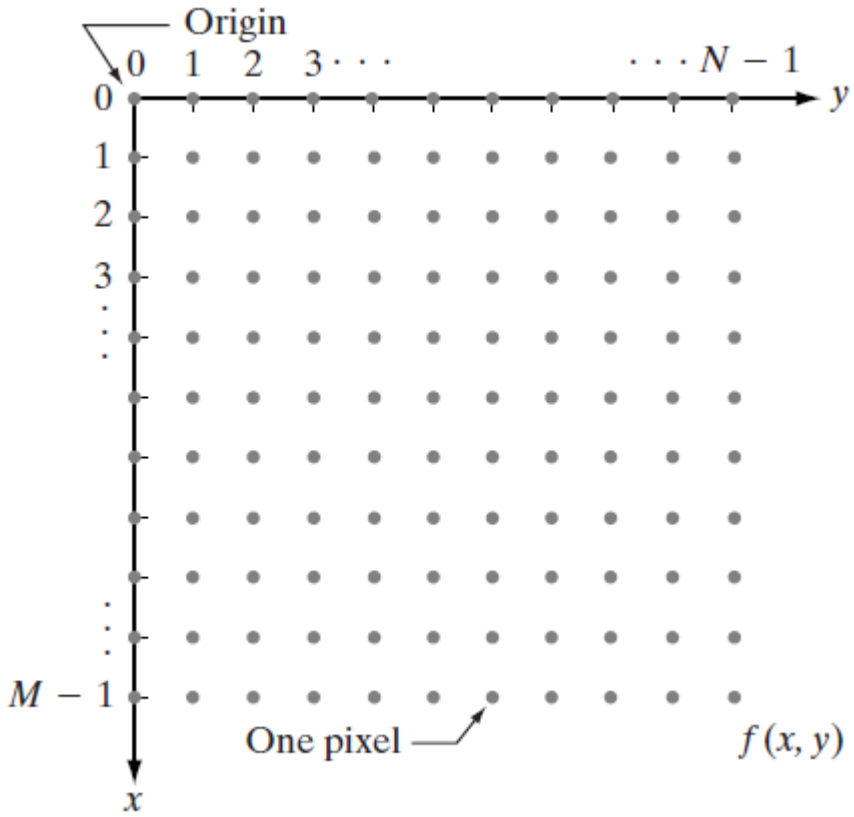
each square is called a pixel (for *picture element*)



Digital 2D image (Grayscale)

Assume that an image f(x, y) is sampled so that the resulting digital image has M rows and N columns.

The values of the coordinates (x, y) now become *discrete* quantities. We will use integer values for these discrete coordinates.



$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}$$

The values of the coordinates at the origin:
(x, y) = (0, 0) for C, C++, Python

Digital 2D image (Grayscale)

- The representation of an M×N image in MATLAB

$$f(x,y)=\begin{bmatrix} f(1,1) & f(1,2) & \dots & f(1,N) \\ f(2,1) & f(2,2) & \dots & f(2,N) \\ \dots & \dots & \dots & \dots \\ f(M,1) & f(M,2) & \dots & f(M,N) \end{bmatrix}$$

The values of the coordinates at the origin:
(x, y) = (1, 1) for MATLAB

Digital 2D image (Grayscale)

Origin at (x, y) = (0, 0) in the textbook and for C, C++, Python, etc.

$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}$$

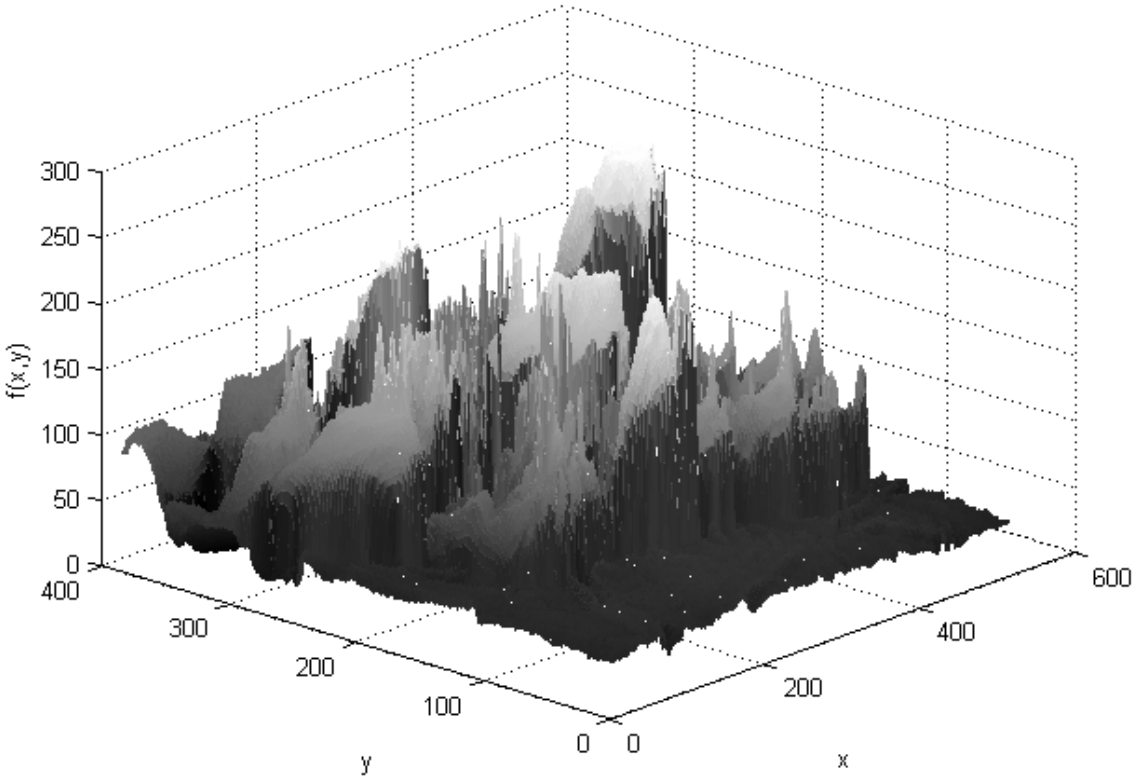
Origin at (x, y) = (1, 1) for MATLAB

$$f(x, y) = \begin{bmatrix} f(1,1) & f(1,2) & \ldots & f(1,N) \\ f(2,1) & f(2,2) & \ldots & f(2,N) \\ \ldots & \ldots & \ldots & \ldots \\ f(M,1) & f(M,2) & \ldots & f(M,N) \end{bmatrix}$$

f is a function that assigns a gray-level value (that is, a real number from the set of real numbers, R) to each distinct pair of coordinates (x, y).

Digital 2D image (Grayscale)





f is a function that assigns a gray-level value (that is, a real number from the set of real numbers, R) to each distinct pair of coordinates (x, y).
$f(x, y)$ represents the «intensity» of light or the «brightness» at pixel location (x, y).
Low values of f correspond to «darker» pixels, high values correspond to «brighter» pixels.

Digital 2D image (Grayscale)

Origin at (x, y) = (0, 0) in the textbook and for C, C++, Python, etc.

$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}$$

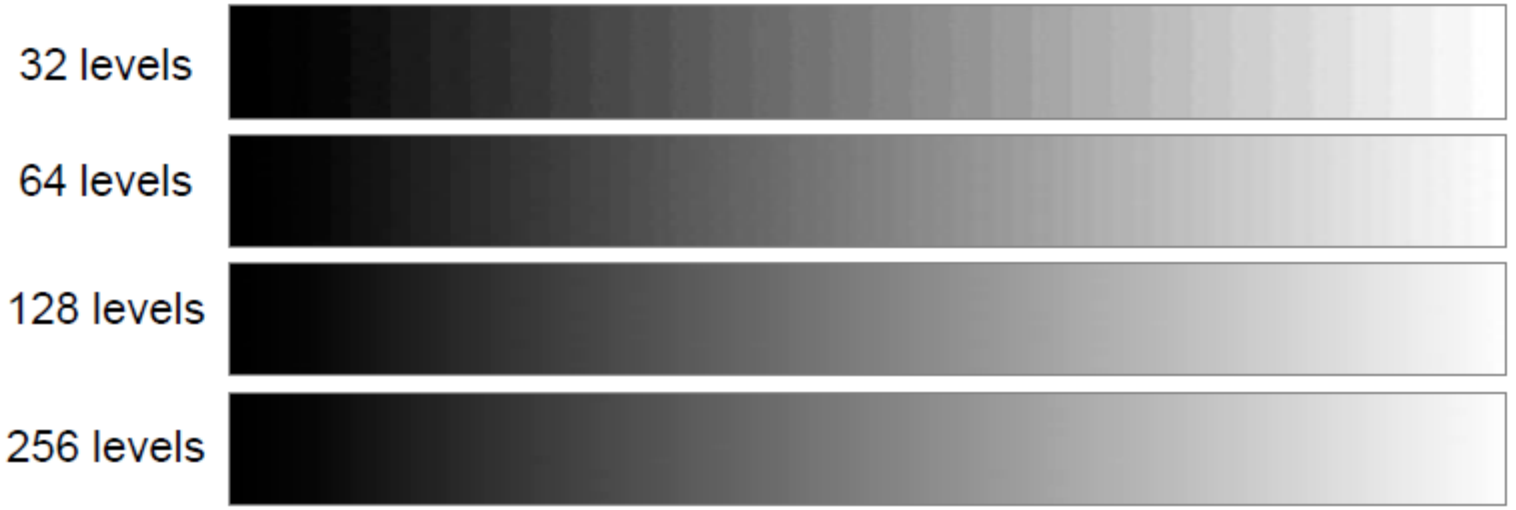
f is a function that assigns a gray-level value (that is, a real number from the set of real numbers, R) to each distinct pair of coordinates (x, y).

If the gray levels also are integers (as usually is the case in this course), Z replaces R, and a digital image then becomes a 2-D function whose coordinates and amplitude values are integers.

The discrete gray levels are equally spaced and that they are integers in the interval [0, L-1].

How many gray levels are required?

- Contouring is most visible for a ramp



- Digital images typically are quantized to 256 gray levels.

Due to processing, storage, and sampling hardware considerations, the number of gray levels, L, typically is an integer power of 2:

$$L = 2^k$$

The discrete levels are equally spaced and that they are integers in the interval $[0, L-1]$. The range of values spanned by the gray scale is called the *dynamic range* of an image.

Digital 2D image (Grayscale)

The number of gray levels, L, typically is an integer power of 2:

$$L = 2^k$$

Given an image of size $M \times N$ ($M \times N$ pixels), with L gray levels, the number of bits required to store the digital image is

$$b = M \times N \times k$$

For example, an image with 256 possible gray-level values is called an 8-bit image ($k = 8$). The number of bits required to store a raw (uncompressed) 8-bit image of size 1024×1024 is

$$b = 1024 \times 1024 \times 8 = 8{,}388{,}608 \text{ bits}$$

MATLAB SESSION

```
% Images are represented as arrays in MATLAB
% Create a 4x6 array to represent a gray level image:

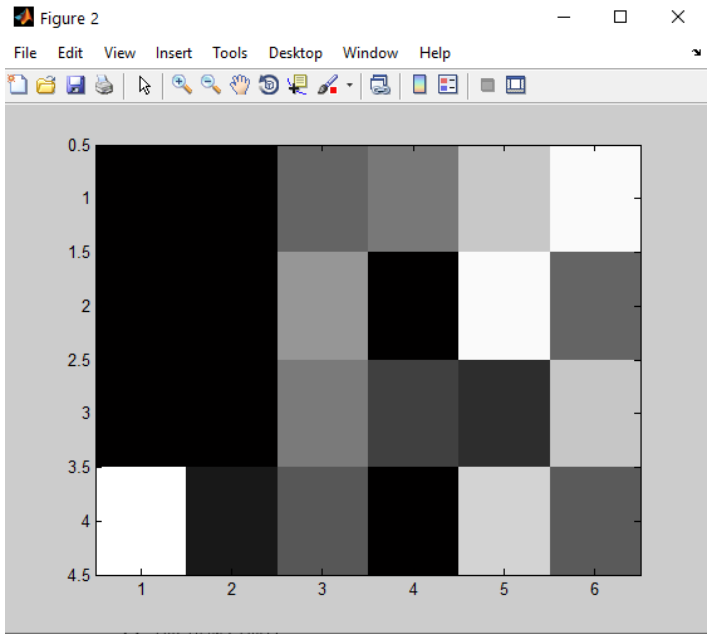
>> Img = [0 0 100 120 200 250;
        0 0 150 0   250 100;
        0 0 122 64  45  198;
        255 24 87 0 211 90];

% Convert to uint8 Unsigned integer 8-bit:

>> Img = uint8(Img);

% Display
>> figure(1);
>> imshow(Img);

% Display (another way)
>> figure(2);
>> image(Img);
>> colormap(gray(256));
```



```
% Clear all the variables from the workspace:
>> clear all
% Close all the figure windows:
>> close all

% Read an image from file
I = imread('landscape.jpg');
% A = imread(FILENAME) reads a grayscale or color image from the file
%    specified by the string FILENAME. If the file is not in the current
%    directory, or in a directory on the MATLAB path, specify the full
%    pathname.


% Display the size of the image
>> size(I)

% Convert to gray scale
>> Igray = rgb2gray(I);

% Display the image
>> figure(1);
>> imshow(Igray);
>> axis on
```

Basic MATLAB Commands

- imread
- size
- whos
- imshow
- imwrite
- im2double
- rgb2gray, im2uint8, im2bw
- img1 = img(1:end-4,:), img1 = img(1:3:end,1:4:end)
- zeros(m,n), ones(m,n)
- rand(m,n), randn(m,n)
- min(min(I1)), max(max(I2))